Engay, Emgee M.

BSIT DA 2-1 W1-Quiz1 Activity Aug. 19, 2026

Tasks:

A. Discuss Briefly:

1. What is an algorithm, and how is it related to the tasks you perform in your daily life?

- An algorithm is a step-by-step process used to finish a task or solve a problem. I use it in my daily life when I prepare for school because I follow different steps before going to school.”

Why is it important for an IT student to learn algorithms?

- It is important for an IT student to learn algorithms because we need to follow the steps to solve a problem in a program and avoid errors in the code.

B. Convert each JavaScript code into a step-by-step algorithm using simple English or pseudocode.

Guide: Describe what the program does from the beginning until the loop stops. You may use numbered steps.

1. for (let i = 1; i < 10; i *= 2) {

   console.log(i);

}

1. Start with i = 1.
2. Check if i is less than 10.
3. Print the value of i.
4. Multiply i by 2.
5. Repeat the steps until i is 10 or more.

Output: 1, 2, 4, 8

2. for (let i = 0; i < 10; i++) {

   console.log(i);

}

1. Start with i = 0.
2. Check if i is less than 10.
3. Print the value of i.
4. Add 1 to i.
5. Repeat until i reaches 10.

Output: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

C. Run both JavaScript codes. Based on the outputs and execution of the two codes, answer the following:

1. What did you observe about the sequence of numbers produced by the first code?

   - I observed that the numbers double each time until they reach the last number. It starts with 1, then becomes 2, 4, and 8.

2. What did you observe about the sequence of numbers produced by the second code?

   - I observed that the numbers increase by 1, starting from 0 until 9.

3. What is the main difference between the way the two loops progress?

   - The difference is that one is multiplied by 2, while the other only adds 1 each time. That is why they produce different sequences of numbers.

4. How does the way an algorithm performs its steps affect its output?

   - The steps of an algorithm affect the output because different steps can give different results.